

# data\_cleaning

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidytext)
```

```
library(textstem)
```

```
## Loading required package: koRpus.lang.en
## Loading required package: koRpus
## Loading required package: syllly
## For information on available language packages for 'koRpus', run
##
##   available.koRpus.lang()
##
## and see ?install.koRpus.lang()
##
## Attaching package: 'koRpus'
##
## The following object is masked from 'package:readr':
##
##   tokenize
```

```
library(rvest)
```

```
##
## Attaching package: 'rvest'
##
## The following object is masked from 'package:readr':
##
##   guess_encoding
```

```
library(qdapRegex)
```

```
##
## Attaching package: 'qdapRegex'
```

```
##
## The following object is masked from 'package:dplyr':
##
##     explain
##
## The following object is masked from 'package:ggplot2':
##
##     %+%
```

```
library(stopwords)
library(tokenizers)
library(here)
```

```
## here() starts at D:/UCSB/Class/F25/PSTAT197A/module-2-claims-data-table-13
```

```
parse_fn <- function(.html){
  page <- tryCatch(
    read_html(.html),
    error = function(e) return(NA)
  )

  if (is.na(page)) return("")

  headers <- page %>%
    html_elements('h1, h2, h3') %>%
    html_text2()

  paragraphs <- page %>%
    html_elements('p') %>%
    html_text2()

  # combine them
  raw_text <- c(headers, paragraphs) %>%
    str_c(collapse = ' ')

  clean_text <- raw_text %>%
    rm_url() %>%
    rm_email() %>%
    str_remove_all('\\') %>%
    str_replace_all(
      paste(c('\\n',
              '[:punct:]',
              'nbsp',
              '[:digit:]',
              '[:symbol:]'),
            collapse = '|'),
      ' '
    ) %>%
    str_replace_all("[a-z]([A-Z])", "\\1 \\2") %>%
    tolower() %>%
    str_replace_all("\\s+", " ")

  return(clean_text)
}
```

```

# function to apply to claims data
parse_data <- function(.df){
  out <- .df %>%
    filter(str_detect(text_tmp, '<!')) %>%
    rowwise() %>%
    mutate(text_clean = parse_fn(text_tmp)) %>%
    unnest(text_clean)
  return(out)
}

# load raw data
load(here("data", "claims-raw.RData"))

clean_df <- parse_data(claims_raw)

save(clean_df, file = here("data", "claims-clean.RData"))

# check cleaned data
clean_df %>%
  select(.id, text_clean) %>%
  slice(1:3)

## # A tibble: 3 x 2
##   .id      text_clean
##   <chr>    <chr>
## 1 url2183 "wayne slim straseske obituary arrangements made by memory book etern~
## 2 url2136 "arrest information for jonathan andrew canola the following official~
## 3 url1586 "find the truth new mugshots in miami dade county fl friday january r~

```